



*Advanced AI-based dynamic Integration
of heterogeneous data for semantic
clusters formalisation in human-centric
intelligent enterprise systems*

Gaining Perspective on Automatic RDF Triple Extraction: models, methods, and pipelines

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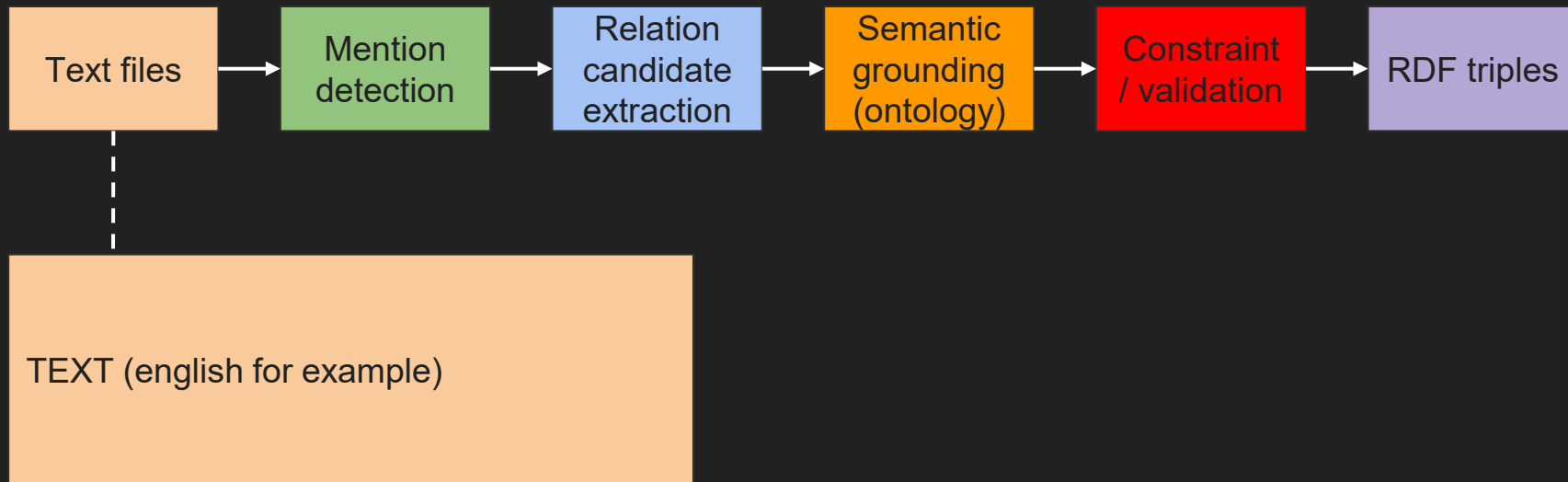
Introduction

Goal : develop and evaluate an ontology-based, supervised pipeline (NLP, ...) for automatic triple extraction from text.



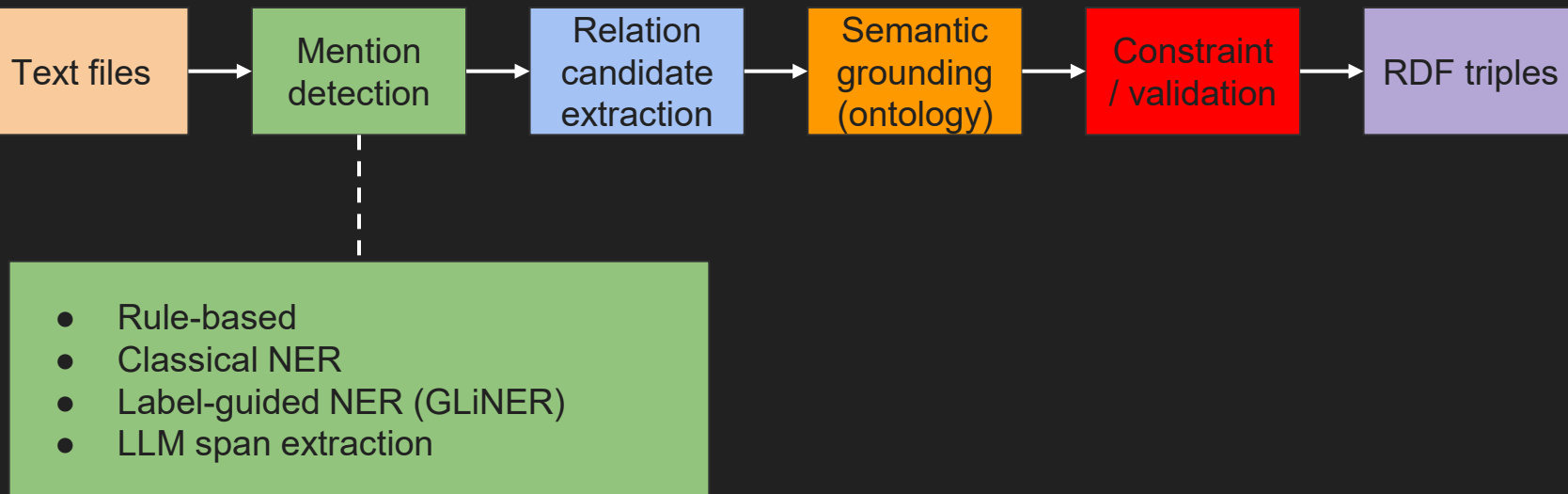
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Text files : plain text



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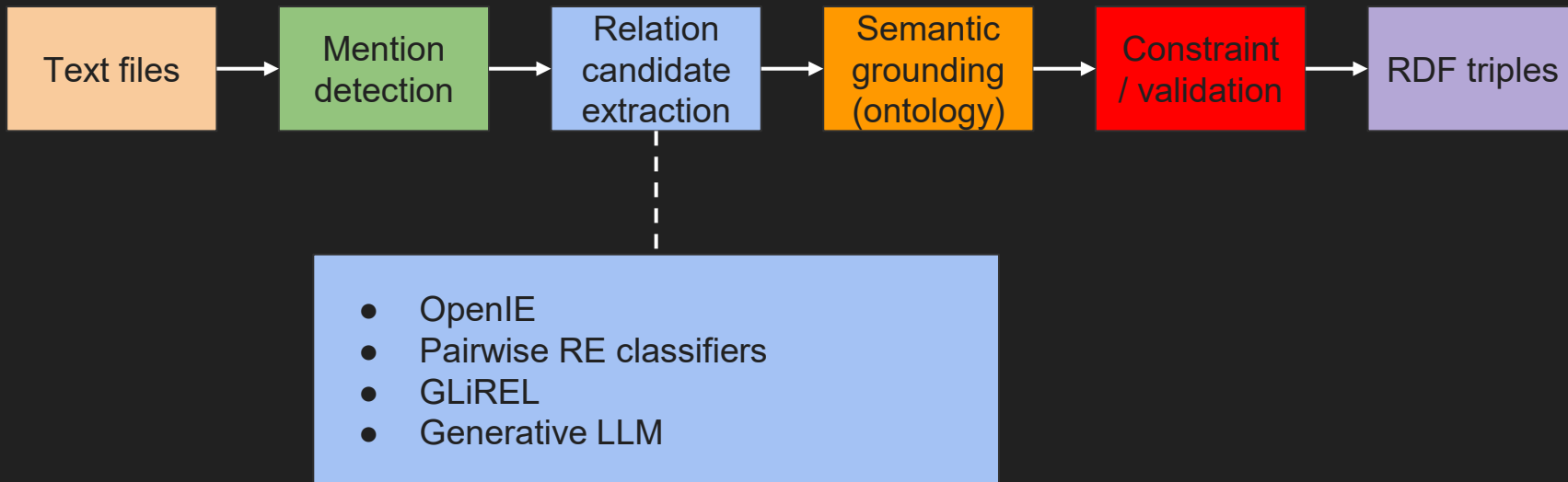
Mention detection : entities, events, attributes, literals (dates, numbers)



Introduction

Relation candidate extraction :

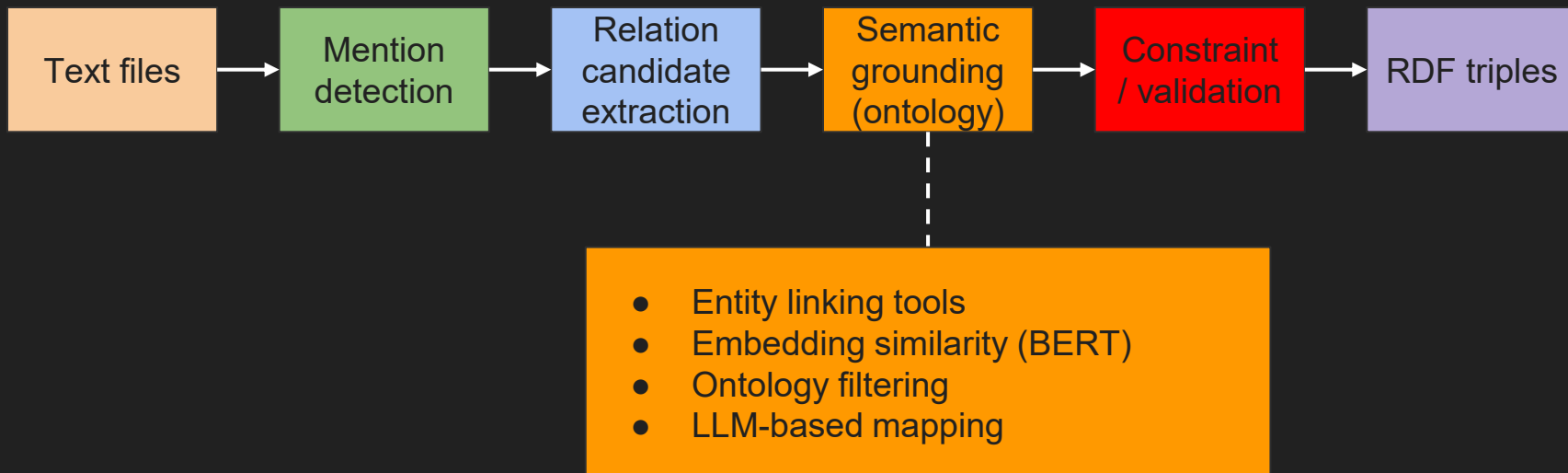
- A. Open (text) vs constrained (ontology-driven)
- B. Local (sentence-level) vs global (document-level)



Introduction

Semantic grounding :

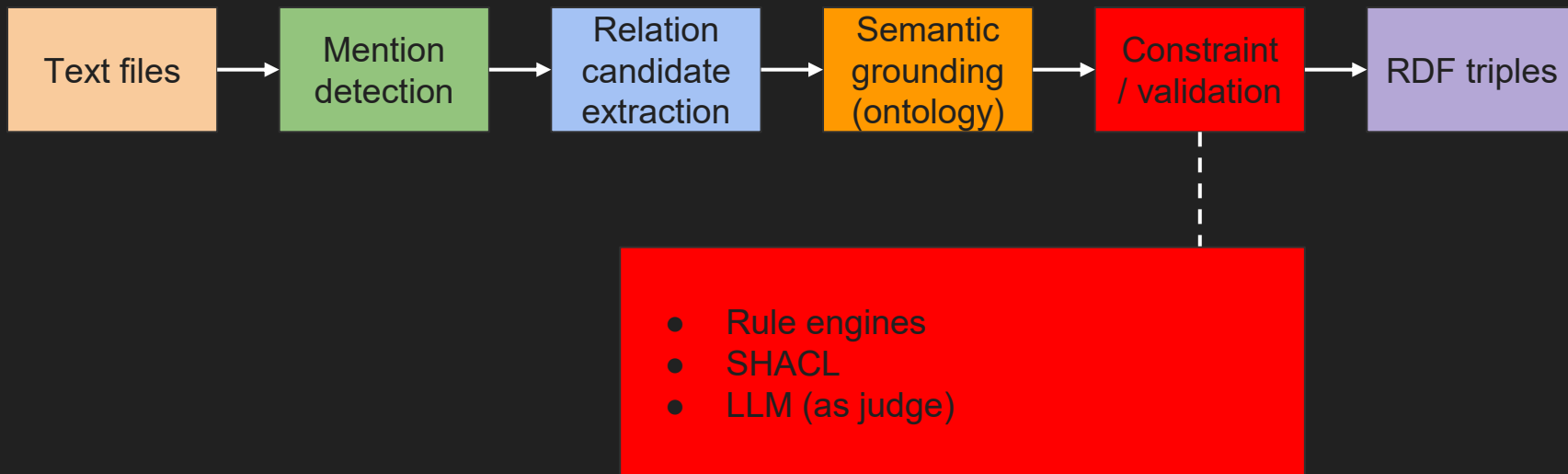
- Entity grounding (mentions $\rightarrow$ URIs)
- Relation grounding (relation $\rightarrow$ ontology property)
- Type grounding (entity $\rightarrow$ class)



Introduction

Constraint / validation :

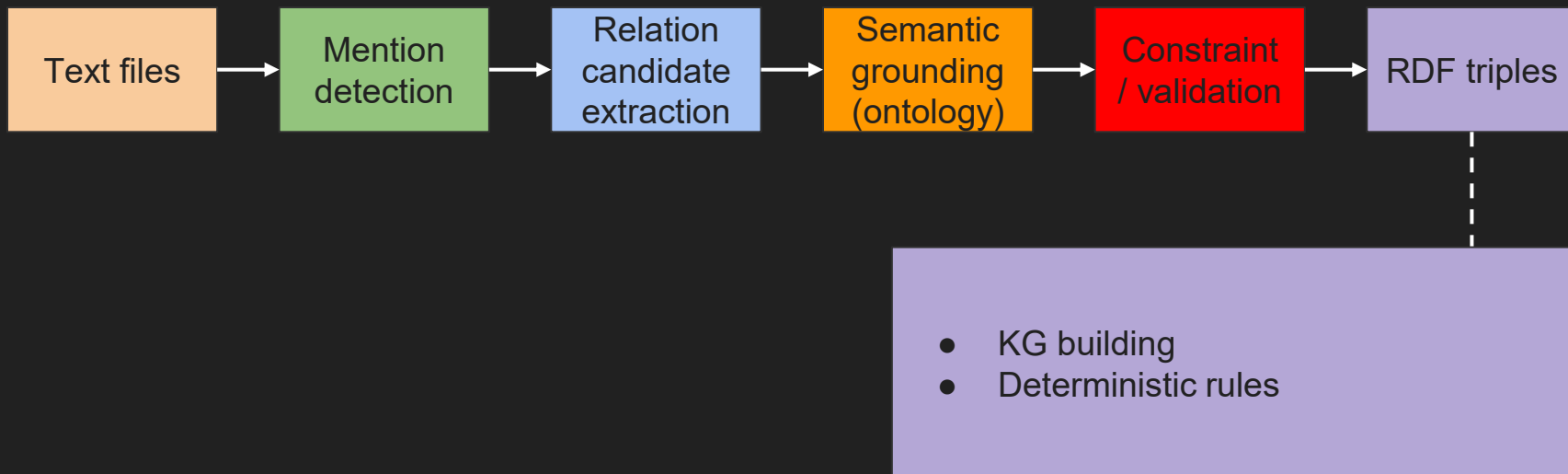
- Domain / range
- Cardinality



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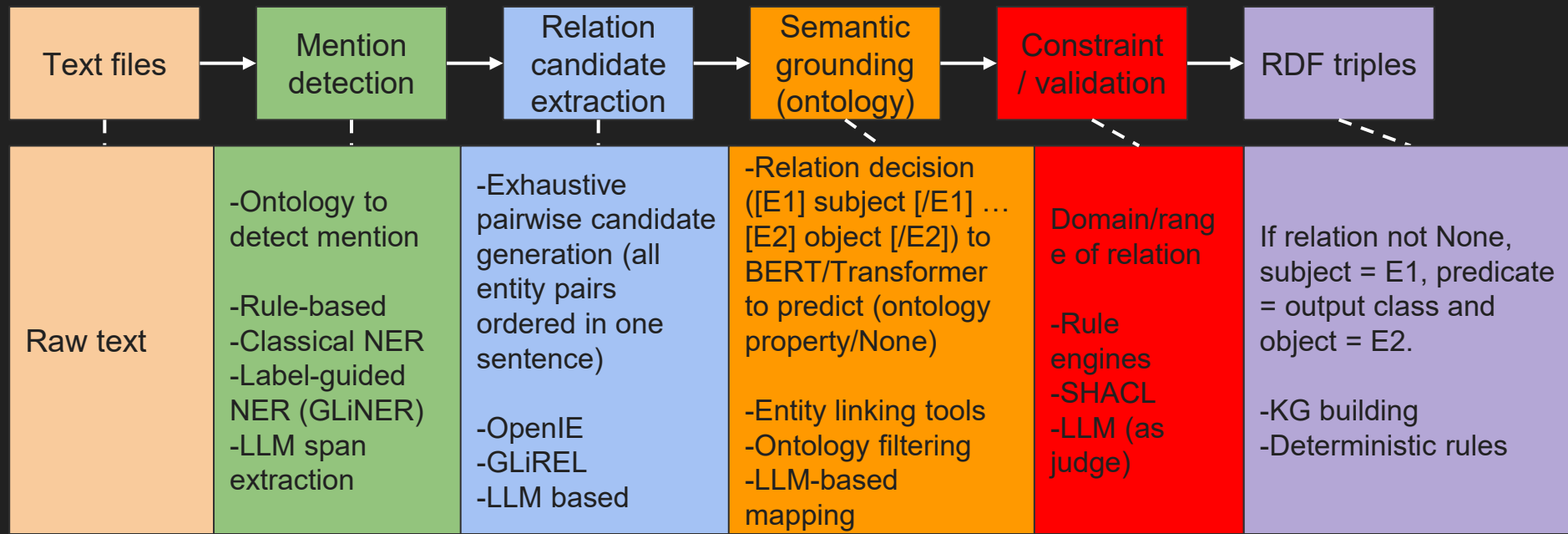
RDF triples formalisation :

- URI normalization, literals, datatypes
- Confidence score



Introduction : what I did

Goal : develop and evaluate an ontology-based, supervised pipeline (NLP, ...) for automatic triple extraction from text.



Introduction : what I did

Problems: Everything depend on the ontology, so if ontology changes, model is no longer well-defined

So it is like a compilation model, an ontology-compiled supervision model because one compiled, it doesn't take into account the changes into my ontology !

GLiNER: good for dynamic ontologies at the entity levels (add class without retraining, rename class) but can't decide/classify relations

GLiREL: schema relation extractor (relation as text labels & external to the model itself)

LLM: very good here but depends prompt and are very costly (computation time, hallucination, latency, non-determinism, ...)

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Benchmark BERT-like models

Goal : better encoder for a BiLM model :

1. DeBERTa-v3 → better contextual comprehension
1. Electra & Albert
1. BERT & RoBERTa

→ in my previous presentation about BERT models, even roberta performed as albert and electra but I didn't try deberta-v3 !

Source GLiNER article:

<https://aclanthology.org/2024.naacl-long.300.pdf>

A Systematic Literature Review on RDF Triple Generation From Natural Language Texts

Key findings:

- Pipeline is generating the RDF triples and then append them into an existing or not KG regarding the requirement of an ontology.
- More and more usage of transformers in 2023, superior accuracy so should be the trend later (still is but usage of LLM in 2026 no ?).
- Before: usage of **OpenIE** (Open Information Extraction), **SRL** (Semantic Role Labeling) + grammatical pattern rules to identify subject-predicate-object structures.
- Now: usage of **NER** (Named Entity Recognition) & **RE** (Relation Extraction): to identify entities and the link between them.
- Majority of research focuses on fully automated approaches to generate KG (most of them are in English)

A Systematic Literature Review on RDF Triple Generation From Natural Language Texts

Key findings:

- Challenges: ambiguity & context, 18% of articles were employed in live
- Trends: RAG where KG serve as structured knowledge to improve the accuracy of the LLM

GLiNER

BERT-models: need to retrain them when there is an update into my ontology and do classification of tokens. (if I add a relation, it won't be able to detect it because wasn't trained on it)

It encodes a global correlation, not a structured relationship.

It only learns that **certain sentence patterns** → **certain relation IDs**.

So BERT is **dataset-driven**, not **ontology-driven**.

GLiNER learns the text and labels but the semantic of it

ontology-defined semantic roles

It is based on POS matching

With GLiNER, the model only identifies semantic arguments, and the ontology decides which triples are allowed.

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GLiNER can detect entity types based on descriptions derived from the ontology

So for each label, provide a description

input: “John works for Google and Meta and lives in California.”

GLiNER

input: “John works for Google and Meta and lives in California.”

John → Person

Google → Organization

Meta → Organization

California → Location

Construction of relations

1) Ontology constraints (domain / range)

worksFor:

domain: Person

range: Organization

2) Linguistic structure

dependency parsing, predicate detection,
argument structure

work

subject → John (Person)

object → Google (Organization)

3) Ontology mapping

work / employ / hired → worksFor

work(John, Google)

work(John, Meta)

live(John, California)

Source GLiNER article: <https://aclanthology.org/2024.naacl-long.300.pdf>

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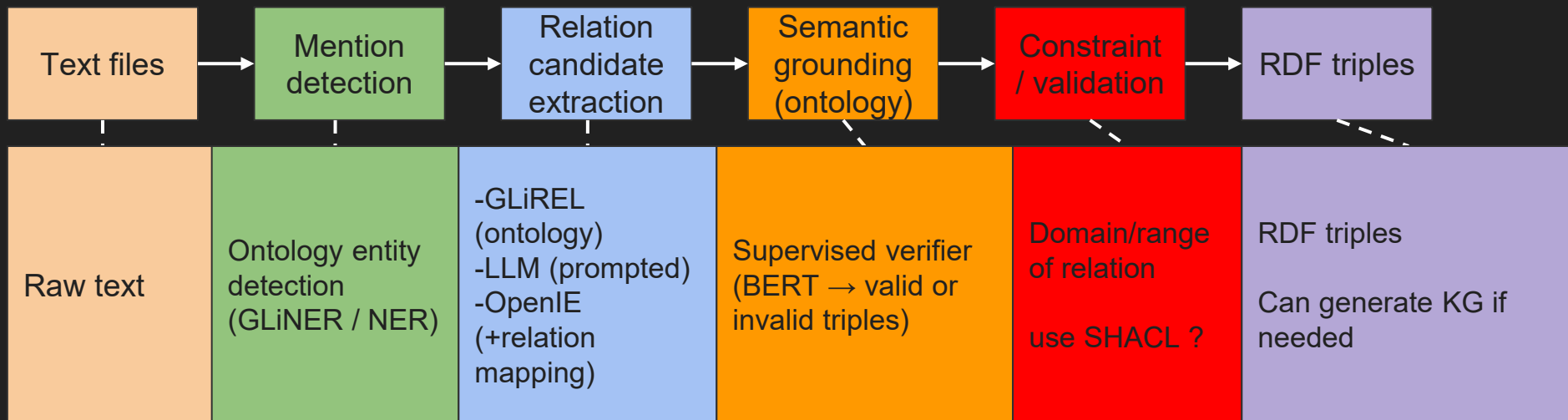
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Architecture 1: Schema Extraction & Supervised verifier

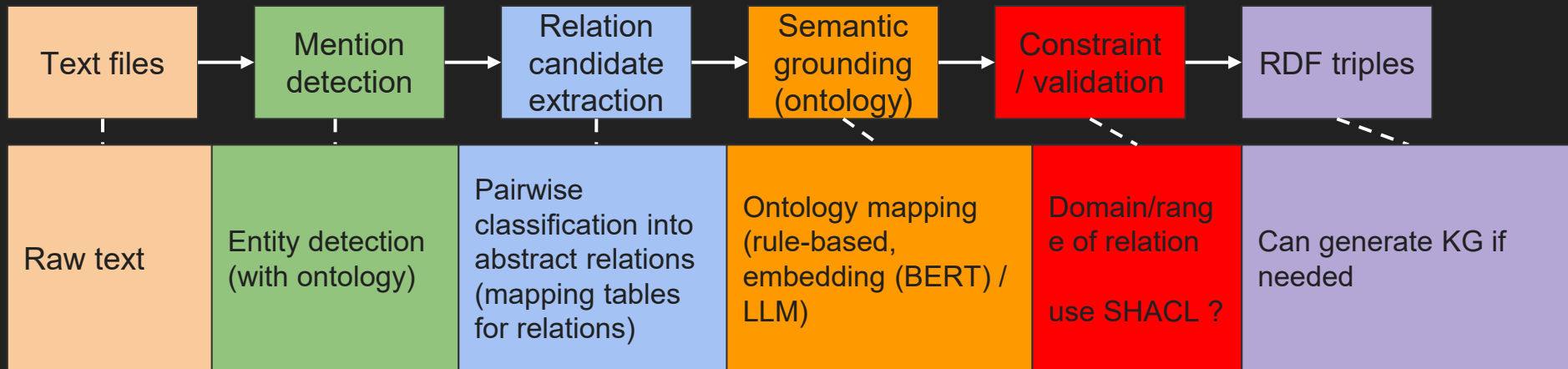
Goal is to dissociate extraction (dynamic) from the final decision (supervised, stable)



- Ontology changes → update the label list
- No dataset rebuild
- BERT is no more the **generator** **BUT** the **verifier** ! It doesn't predict relation anymore but verify them !

Architecture 2: abstract relation & ontology mapping

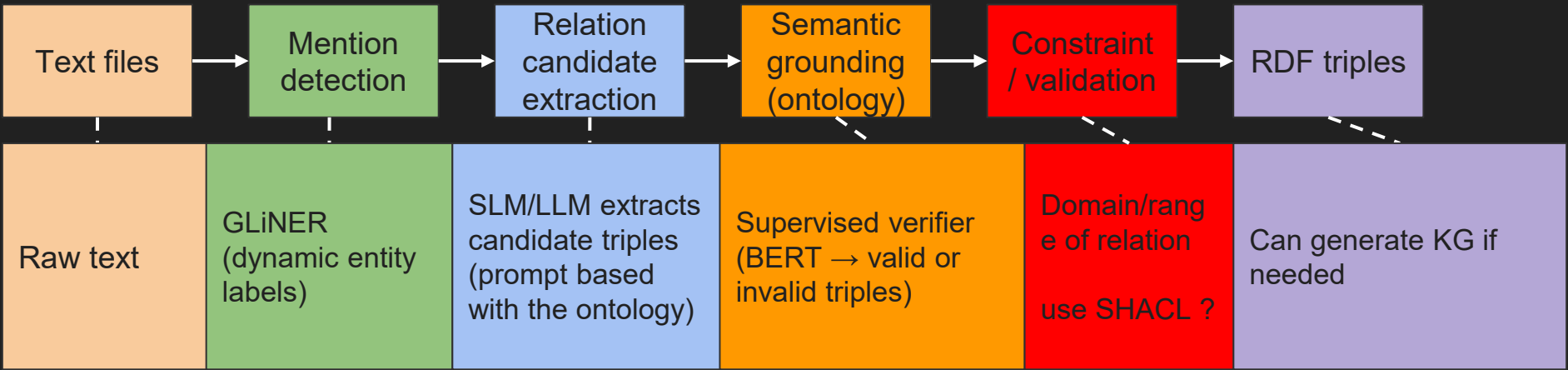
Stabilize learning by predicting **abstract relationships**, independent of concrete ontology (same goal as architecture 1 = dynamic)



- Add ontology predication == update the mapping table ?
- Similar to what I did but try to be dynamic

Architecture 3: ontology with LLM & Supervised verifier

Maximize flexibility and semantic coverage with LLMs, while maintaining a supervised security nest.



- Very flexible and ontology based (so should prevent the hallucination problem)
- Cons: depend on the quality of the prompt and LLM are costly (computation, latency, ...) so maybe SLM ?

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Thank you for your attention !

Questions ?